



May 2025

Duration: 3 Hours

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What do National Building Codes and Rules primarily focus on?**
 - Aesthetic considerations
 - Structural design only
 - Safety and standards
 - Interior decoration
- What is the primary basis for the classification of buildings?**
 - Architectural style
 - Height
 - Location
 - Occupancy
- What is the purpose of building byelaws?**
 - Aesthetic enhancements
 - Safety regulations
 - Architectural style
 - Historical preservation
- What does FAR stand for in architecture?**
 - Floor Area Renovation
 - Floor Area Ratio
 - Farmland Area Ratio
 - Functional Architecture Rating
- In architectural design, what does the term "elevation" typically refer to?**
 - A front view of a building
 - The foundation design
 - The floor plan
 - The site plan
- In architectural planning, what does the term "orientation" refer to?**
 - Location on the map
 - Interior design layout
 - Alignment with cardinal directions
 - Building height
- What is the main purpose of site plans in architectural presentation**
 - Showcasing interior design
 - Presenting building elevations
 - Displaying the site layout
 - Highlighting structural details
- In architectural design, what does the term "plan view" typically refer to?**
 - Front view
 - Top view
 - Side view
 - Cross section view
- What is the primary importance of Vaastu Shastra in modern times?**
 - Aesthetic enhancement
 - Cultural preservation
 - Spiritual harmony
 - Technological advancements
- In Vaastu Shastra, which element is associated with the fire energy**
 - Earth (Prithvi)
 - Fire (Agni)
 - Water (Jal)
 - Air (Vayu)
- According to Vaastu Shastra, what is a key consideration when selecting a site?**
 - Site popularity
 - Site direction and slope
 - Site accessibility
 - Site proximity to shopping centers
- How does Vaastu Shastra view the impact of neighboring structures on site selection?**
 - Irrelevant to site selection
 - Provides additional security
 - Can influence energy flow
 - Enhances aesthetic appeal

13. According to Vaastu Shastra, which color is associated with positive energy
 A) Black B) White
 C) Red D) Pink
14. In Vaastu Shastra, what is the significance of the northeast direction in room layout
 A) Ideal for kitchens B) Ideal for pooja room
 C) Best for bedrooms D) Ideal for toilets
15. According to Vaastu Shastra, how should lighting be arranged in a room
 A) Harsh and direct lighting B) Natural and soft lighting
 C) Dim and colorful lighting D) Fluorescent lighting
16. What is a common remedy suggested by Vaastu Shastra to rectify negative energy in a space?
 A) Use of metallic furniture B) Placement of mirrors
 C) Burning scented candles D) Hanging wind chimes
17. Which of the following is a key principle of solar passive cooling?
 A) Maximizing heat gain during the day B) Reducing heat gain and enhancing ventilation
 C) Using artificial cooling methods D) Increasing the use of non-renewable energy sources
18. Which of the following is NOT a low-energy cooling technique?
 A) Evaporative cooling B) Cross ventilation
 C) Geothermal heating D) Night flushing
19. Which solid waste management practice is most sustainable for green buildings?
 A) Open dumping B) Landfilling
 C) Recycling and composting D) Incineration without emission control
20. Which feature is commonly integrated into urban green buildings?
 A) Solar panels and green roofs B) High-energy lighting
 C) Fossil fuel-based heating D) Minimal vegetation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the purpose of building byelaws in urban planning and development. Provide two specific examples of building byelaws and their impact on construction.	8	L*2	1	1
	b) Develop a line diagram of an East facing two-bed room residential building with a site area of 20m*15m by assuming suitable dimensions for each component.	8	L3	2	2
2.	a) Explain the term "Floor Area Ratio (FAR)" and discuss its importance in building design. Provide a numerical example to demonstrate how FAR is calculated.	8	L2	1	1
	b) Develop the line diagram of a college canteen building as per the standards.	8	L3	2	2
3.	a) Explain the classification of building and provide examples of different types of buildings based on occupancy.	8	L2	1	1
	b) Plan the line diagram of a commercial complex buildings as per the standards.	8	L3	2	2
Unit – II					
4.	a) Explain the principles of choosing a site based on Vaastu Shastra, considering factors like direction, shape, and slope.	8	L2	3	1
	b) What are the guidelines for the placement of rooms and doors based on Vaastu Shastra?	8	L1	4	1
5.	a) How does the individual elements in Vaastu Shastra contribute to the overall harmony and balance in a building.	8	L2	3	1
	b) Develop the bedroom layout and interior decor plan based on Vaastu Shastra principles.	8	L3	4	2

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| 6. a) | Explain the impact of neighboring structures and environmental factors on site selection. | 8 | L1 | 3 | 1 |
| b) | Explain the principles of lighting, ventilation, and space utilization according to Vaastu Shastra. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Explain the concept of solar passive cooling in buildings. How does it help in reducing energy consumption? | 8 | L2 | 5 | 1 |
| b) | List the importance of water utilization planning in green building design. | 8 | L1 | 5 | 1 |
| 8. a) | What are the design considerations for incorporating solar energy in buildings? Discuss with examples. | 8 | L1 | 5 | 1 |
| b) | Explain the advantages and challenges of using green composites in the building industry. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
